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## 6 1. Introduction

7 Thick plate theory with Mindlin-Reissner hyper..., or said the First-order  
8 Strain Deformation Theory (FSDT), is commonly used in finite element struc-  
9 ture analysis. This theory added the independent rotation degrees of freedoms  
10 (DOFs), and unlimited the usage of  $C^0$  continuous shape functions, that should  
11 be required in Kirchhoff-Love hyper.... However,

## 12 2. Reissner–Mindlin plate formulation

### 13 2.1. Governing equations for Mindlin plate

14 Consider a thick plate  $\Omega \times [-\frac{t}{2}, \frac{t}{2}]$  with its mid-surface  $\Omega$  and thickness  $t$  in  
15 a Cartesian coordinate system, as shown in Figure, where  $x_\alpha$ ,  $\alpha = 1, 2$  are the  
16 in-plane coordinates on  $\Omega$  and  $x_3$  is the coordinate for the thickness direction.  
17 In accordance with Reissner-Mindlin hypothesis and inconsideration of the in-  
18 plane displacement of  $\Omega$ , the displacement  $u_i$ ,  $i = 1, 2, 3$  are given as follows:

$$\begin{cases} u_\alpha(\mathbf{x}) = -x_3\varphi_\alpha(x_1, x_2) \\ u_3(\mathbf{x}) = w(x_1, x_2) \end{cases}, \quad \alpha = 1, 2 \quad (1)$$

19 where  $w$  stands for the deflection and  $\varphi_\alpha$  presents  $\alpha$ th direction component for  
20 rotation  $\boldsymbol{\varphi}$ . The corresponding infinitesimal strain components  $\varepsilon_{ij}$  are listed as  
21 follows:

$$\begin{cases} \varepsilon_{\alpha\beta} = \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}) = x_3\kappa_{\alpha\beta} \\ \varepsilon_{\alpha 3} = \frac{1}{2}\gamma_\alpha \\ \varepsilon_{33} = 0 \end{cases} \quad (2)$$

22 in which  $\kappa_{\alpha\beta}$  and  $\gamma_\alpha$  are the components for the in-plane curvature  $\boldsymbol{\kappa}$  and shear  
23 strain  $\boldsymbol{\gamma}$  respectively, the  $\boldsymbol{\kappa}$  and  $\boldsymbol{\gamma}$  are defined as:

$$\boldsymbol{\kappa} = -\frac{1}{2}(\nabla\boldsymbol{\varphi} + \boldsymbol{\varphi}\nabla) \quad (3)$$

$$\boldsymbol{\gamma} = \nabla w - \boldsymbol{\varphi} \quad (4)$$

24 According to the isotropic Mindlin plate theory [1], the stress components  
25  $\sigma_{ij}$  are given as follows:

$$\begin{cases} \sigma_{\alpha\beta} = D_{\alpha\beta\gamma\eta}\varepsilon_{\gamma\eta} \\ \sigma_{\alpha 3} = G\gamma_\alpha \\ \sigma_{33} = 0 \end{cases} \quad (5)$$

26 where  $D_{\alpha\beta\gamma\eta}$  is denoted as the component for the fourth-order in-plane consti-  
27 tutive tensor  $\mathbf{D}$  and  $G$  is shear modulus, and they can be evaluated by Young's

modulus  $E$ , Poisson's ratio  $\nu$  as:

$$\mathbf{D} = \frac{E}{1-\nu^2} \left( \nu \mathbf{1} \otimes \mathbf{1} + \frac{1}{2}(1-\nu) \mathbf{I} \right) \quad (6)$$

$$G = \frac{E}{2(1+\nu)} \quad (7)$$

in which  $\mathbf{1} = \delta_{\alpha\beta} \mathbf{e}_i \otimes \mathbf{e}_j$  and  $\mathbf{I} = \delta_{\alpha\gamma} \delta_{\beta\eta} + \delta_{\alpha\eta} \delta_{\beta\gamma} \mathbf{e}_\alpha \otimes \mathbf{e}_\beta \otimes \mathbf{e}_\gamma \otimes \mathbf{e}_\eta$  represent the second- and fourth-order identity tensors, and  $\delta_{\alpha\beta}$  is Kronecker delta function. Further integrating the stress components through the thickness direction, we can obtain the moment resultant  $\mathbf{m}$  and shear force resultant  $\mathbf{q}$ , given as follows:

$$\mathbf{m} = \int_{-t/2}^{t/2} \sigma_{\alpha\beta} dx_3 \mathbf{e}_\alpha \otimes \mathbf{e}_\beta = \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} \quad (8)$$

$$\mathbf{q} = k \int_{-t/2}^{t/2} \sigma_{\alpha 3} dx_3 \mathbf{e}_\alpha = k G t \boldsymbol{\gamma} \quad (9)$$

where  $k$  presents the shear correction factor that is chosen as  $k = \frac{4}{5}$  in this work. It should be noted that, the geometric parameter for the moment resultant has two more order power of thickness than shear force resultant. As the thickness reducing, the parameter for the moment resultant will much less than that for shear force resultant, for a specific external force, leading to a constraint of shear strain trending to be zero, i.e.  $t \rightarrow 0, \boldsymbol{\gamma} \rightarrow \mathbf{0}$ .

Consequently, the governing equations for Reissner-Mindlin plate can be expressed as follows:

$$\begin{cases} \nabla \cdot \mathbf{m} - \mathbf{q} = \mathbf{0} & \text{in } \Omega \\ \nabla \cdot \mathbf{q} + \bar{p} = 0 & \text{in } \Omega \\ \mathbf{q} \cdot \mathbf{n} = \bar{q}_n & \text{on } \Gamma_q \\ \mathbf{m} \cdot \mathbf{n} = \bar{\mathbf{m}}_n & \text{on } \Gamma_m \\ \boldsymbol{\varphi} = \bar{\boldsymbol{\varphi}} & \text{on } \Gamma_\varphi \\ w = \bar{w} & \text{on } \Gamma_w \end{cases} \quad (10)$$

where  $\bar{p}$  is the transverse load,  $\bar{q}_n$  and  $\bar{\mathbf{m}}$  are the prescribed shear force and moment on the natural boundary  $\Gamma_q$  and  $\Gamma_m$  respectively,  $\bar{w}$  and  $\bar{\boldsymbol{\varphi}}$  are the prescribed deflection and rotation on the essential boundary  $\Gamma_w$  and  $\Gamma_\varphi$  respectively, and  $\mathbf{n}$  is the unit outward normal vector on the boundary. All the boundary segments satisfy the following relationship:

$$\Gamma_q \cup \Gamma_m = \Gamma_w \cup \Gamma_\varphi = \Gamma, \quad \Gamma_w \cap \Gamma_\varphi = \Gamma_q \cap \Gamma_m = \emptyset \quad (11)$$

and  $\Gamma$  is the boundary of  $\Omega$ .

## 2.2. Mixed-formulation

The thick plate Galerkin formulation generally use the three-variable weak form to immigrate the locking issue. To obtain the corresponding Galerkin weak

form, the complementary energy functional is given by:

$$\begin{aligned}\Pi(\mathbf{u}, \mathbf{q}) = & \int_{\Omega} \frac{t^3}{24} \boldsymbol{\kappa} : \mathbf{D} : \boldsymbol{\kappa} d\Omega - \int_{\Omega} \frac{1}{2kGt} \mathbf{q} \cdot \mathbf{q} d\Omega - \int_{\Omega} \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega \\ & + \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma + \int_{\Gamma_m} \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{q}_n) d\Gamma \\ & - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega\end{aligned}\quad (12)$$

where  $\mathbf{u}$  stands for a displacement set containing deflection  $w$  and rotation  $\boldsymbol{\varphi}$ , i.e.  $\mathbf{u} = \{w, \boldsymbol{\varphi}\}$ . Introducing the standard variational operation herein leads to the following Galerkin weak form: find  $w \in W$ ,  $\boldsymbol{\varphi} \in \Phi$ ,  $\mathbf{q} \in Q$ ,

$$\begin{aligned}a_{\varphi}(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) + b_{\varphi}(\delta\boldsymbol{\varphi}, \mathbf{q}) &= f_{\varphi}(\delta\boldsymbol{\varphi}), \quad \forall \delta\boldsymbol{\varphi} \in \Phi \\ b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\ a_q(\delta\mathbf{q}, \mathbf{q}) + b_q(\delta\mathbf{q}, \mathbf{u}) &= f_q(\delta\mathbf{q}), \quad \forall \delta\mathbf{q} \in Q\end{aligned}\quad (13)$$

where the deflection space  $W$ , rotation space  $\Phi$  and shear force space  $Q$  are defined as:

$$W := \{\} \quad (14)$$

$$\Phi := \{\} \quad (15)$$

$$Q := \{\} \quad (16)$$

and the bilinear forms and linear forms in Eq. (13) are detailed as follows:

$$a_{\varphi}(\delta\boldsymbol{\varphi}, \boldsymbol{\varphi}) = \int_{\Omega} \delta\boldsymbol{\kappa} : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa} d\Omega \quad (17)$$

$$a_q(\delta\mathbf{q}, \mathbf{q}) = - \int_{\Omega} \frac{1}{2kGt} \delta\mathbf{q} \cdot \mathbf{q} d\Omega \quad (18)$$

$$b_q(\mathbf{q}, \mathbf{u}) = b_w(w, \mathbf{q}) + b_{\varphi}(\boldsymbol{\varphi}, \mathbf{q}) \quad (19)$$

$$b_{\varphi}(\boldsymbol{\varphi}, \mathbf{q}) = - \int_{\Omega} \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega \quad (20)$$

$$\begin{aligned}b_w(w, \mathbf{q}) &= \int_{\Gamma_q} w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} w \nabla \cdot \mathbf{q} d\Omega \\ &= \int_{\Omega} \nabla w \cdot \mathbf{q} d\Omega - \int_{\Gamma_w} w \mathbf{q} \cdot \mathbf{n} d\Gamma\end{aligned}\quad (21)$$

$$f_{\varphi}(\boldsymbol{\varphi}) = - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \quad (22)$$

$$f_w(w) = \int_{\Gamma_q} w \bar{q}_n d\Gamma + \int_{\Omega} w \bar{p} d\Omega \quad (23)$$

$$f_q(\mathbf{q}) = - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \quad (24)$$

Furthermore, we introduce the approximation schemes to discrete these three variables, namely  $w_h$ ,  $\varphi_h$ ,  $\mathbf{q}_h$ , these approximants are represented by shape functions  $N_I^w$ ,  $N_K^\varphi$ ,  $N_R^q$  and their corresponding nodal coefficients  $w_I$ ,  $\varphi_K$ ,  $\mathbf{q}_R$  respectively, yields:

$$\begin{cases} w_h(\mathbf{x}) = \sum_{I=1}^{n_w} N_I^w(\mathbf{x}) w_I \\ \varphi_h(\mathbf{x}) = \sum_{K=1}^{n_\varphi} N_K^\varphi \varphi_K \\ \mathbf{q}_h(\mathbf{x}) = \sum_{R=1}^{n_q} N_R^q(\mathbf{x}) \mathbf{q}_R \end{cases} \quad (25)$$

Substituting Eq. (25) back into Eq. (13) can obtain the Galerkin weak formulation: find  $w_h \in W_h$ ,  $\varphi_h \in \Phi_h$ ,  $\mathbf{q}_h \in Q_h$ ,

$$a(\delta\varphi_h, \varphi_h) + b_\varphi(\delta\varphi_h, \mathbf{q}_h) = f_\varphi(\delta\varphi_h), \quad \forall \delta\varphi_h \in \Phi_h \quad (26a)$$

$$b_w(\delta w_h, \mathbf{q}_h) = f_w(\delta w_h), \quad \forall \delta w_h \in W_h \quad (26b)$$

$$b(\delta\mathbf{q}_h, \mathbf{u}_h) = f_q(\delta\mathbf{q}_h), \quad \forall \delta\mathbf{q}_h \in Q_h \quad (26c)$$

where  $\mathbf{u}_h = \{w_h, \varphi_h\}^T$ . The related approximation spaces are defined as:

$$W_h := \text{span}\{N_I^w\}_{I=1}^{n_w} \quad (27)$$

$$\Phi_h := \text{span}\{N_K^\varphi\}_{K=1}^{n_\varphi} \quad (28)$$

$$Q_h := \text{span}\{N_R^q\}_{R=1}^{n_q} \quad (29)$$

FIXME: approximation types

$$\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} & \mathbf{K}^{q\varphi T} \\ \mathbf{0} & \mathbf{0} & \mathbf{K}^{qwT} \\ \mathbf{K}^{q\varphi} & \mathbf{K}^{qw} & \mathbf{K}^{qq} \end{bmatrix} \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \\ \mathbf{d}^q \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi \\ \mathbf{f}^w \\ \mathbf{f}^q \end{Bmatrix} \quad (30)$$

$$\left( \underbrace{\begin{bmatrix} \mathbf{K}^{\varphi\varphi} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{K}^{\text{bending}}} + \underbrace{\begin{bmatrix} \tilde{\mathbf{K}}^{\varphi\varphi} & \tilde{\mathbf{K}}^{\varphi w} \\ \tilde{\mathbf{K}}^{w\varphi} & \tilde{\mathbf{K}}^{ww} \end{bmatrix}}_{\tilde{\mathbf{K}}^{\text{shearing}}} \right) \begin{Bmatrix} \mathbf{d}^\varphi \\ \mathbf{d}^w \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}^\varphi - \tilde{\mathbf{f}}^\varphi \\ \mathbf{f}^w - \tilde{\mathbf{f}}^w \end{Bmatrix} \quad (31)$$

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$$\tilde{\mathbf{K}}^{\varphi\varphi} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{q\varphi} \quad (32)$$

$$\tilde{\mathbf{K}}^{\varphi w} = -\mathbf{K}^{q\varphi T} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (33)$$

$$\tilde{\mathbf{K}}^{ww} = -\mathbf{K}^{qwT} (\mathbf{K}^{qq})^{-1} \mathbf{K}^{qw} \quad (34)$$

**Remark 1.** As shown in Eq. (26), there are two constraints in thick plate mixed formulation. One, as listed in Eq. (26b), is a constraint mainly regarding the shear force resultant on natural boundary  $\Gamma_q$  and its divergence in entire

69 domain. The other is deflection essential boundary condition  $\Gamma_w$  and the shear  
 70 strain enforcement in whole domain, as listed in Eq. (26c). However, for the  
 71 most of pioneer works, only consider the shear strain constraint, and that is not  
 72 sufficient for a satisfaction result.

73 **Remark 2.**

### 74 3. Optimal constraint ratio for shear locking

75 The previous section has been mentioned that the shear locking is consisted  
 76 of two parts, shear force locking and shear strain locking. The corresponding  
 77 LBB conditions are listed as follows:

- 78 • Shear force locking LBB condition:

$$\inf_{w_h \in W_h} \sup_{\mathbf{q}_h \in Q_h} \frac{|b_w(w_h, \mathbf{q}_h)|}{\|w_h\|_W \|\mathbf{q}_h\|_Q} \geq \beta_q \quad (35)$$

- 79 • Shear strain locking LBB condition:

$$\inf_{\mathbf{q}_h \in Q_h} \sup_{\mathbf{u}_h \in W_h \times \Phi_h} \frac{|b(\mathbf{q}_h, \mathbf{u}_h)|}{\|\mathbf{q}_h\|_Q \|\mathbf{u}_h\|_{W \times \Phi}} \geq \beta_\gamma \quad (36)$$

80 where,  $\beta_q$  and  $\beta_\gamma$  are LBB coefficients. The norms in the above LBB conditions  
 81 are defined as follows:

$$\|w_h\|_W^2 = \int_{\Omega} \delta w_h k G h w_h d\Omega \quad (37)$$

$$\|\mathbf{q}_h\|_Q^2 = \int_{\Omega} \frac{\delta \mathbf{q}_h \cdot \mathbf{q}_h}{k G t} d\Omega \quad (38)$$

$$\|\boldsymbol{\varphi}_h\|_{\Phi}^2 = \int_{\Omega} \delta \boldsymbol{\kappa}_h : \frac{t^3}{12} \mathbf{D} : \boldsymbol{\kappa}_h d\Omega \quad (39)$$

$$\|\mathbf{u}_h\|_{W \times \Phi}^2 = \|w_h\|_W^2 + \|\boldsymbol{\varphi}_h\|_{\Phi}^2 \quad (40)$$

82 Next, we will focus on these two LBB conditions respectively and employ the  
 83 complete-polynomial-based estimate method [2] to investigate the relationship  
 84 between their LBB coefficients  $\beta_q$  and  $\beta_\gamma$  and the degree of freedoms (DOFs)  
 85 for variables  $w_h$ ,  $\boldsymbol{\varphi}_h$  and  $\mathbf{q}_h$ . The detailed path for the complete-polynomial-  
 86 based estimate method is listed in Appendix A. Consequently, the optimal ratio  
 87 of degrees of freedoms (DOFs) between variables can be revealed based on these  
 88 two estimators.

#### 89 3.1. Estimator for $\beta_q$

90 The coefficient  $\beta_q$  controls the behavior of shear force locking in Eq. (26b),  
 91 the detailed enforcement is listed as follows:

$$\begin{aligned} & b_w(\delta w_h, \mathbf{q}_h) - f_w(\delta w_h) = (\delta w_h, \mathcal{W} \mathbf{q}_h) - f_w(\delta w_h) \\ & = \int_{\Gamma_q} \delta w_h (\mathbf{q}_h \cdot \mathbf{n} - \bar{q}_n) d\Gamma - \int_{\Omega} \delta w_h (\nabla \cdot \mathbf{q}_h + \bar{p}) d\Omega \\ & = 0 \end{aligned} \quad (41)$$

where, the constraints includes two parts, one is the equilibrium equation between shear force resultant and transverse load in  $\Omega$ , the other is the natural shear force boundary condition on  $\Gamma_q$ . In accordance with the general form of completed-polynomial-based estimator mentioned in Appendix A, the operator  $\mathcal{W} : Q \rightarrow W$  is a specific instance of  $\mathcal{P}$  in Theorem.1, and is defined as:

$$\mathcal{W} = \begin{cases} \mathcal{W}^\Omega = -\nabla \cdot & \text{in } \Omega \\ \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}, \mathcal{W}^{\Gamma_q} = \mathbf{n} \cdot & \text{on } \Gamma_q \end{cases} \quad (42)$$

where, based on the trace theorem [],  $\mathcal{T}^{\Gamma_q} : Q \rightarrow Q|_{\Gamma_q}$  is a trace operator mapping the shear force  $\mathbf{q}$  from domain  $\Omega$  to the boundary  $\Gamma_q$ , while the parametric variable of  $\mathbf{q}$  on boundary can be represented using  $s$ , i.e.  $\mathbf{q}(s) \in Q|_{\Gamma_q}$ . According to Theorem.1, the completed-polynomial-based estimator of  $\beta_q$  has the following form:

$$\beta_q \leq \inf_{Q' \subset Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q} \sup_{\mathbf{q} \in Q'} \frac{\|\mathcal{W}\mathbf{q}\|_W}{\|\mathbf{q}\|_Q} + O(h) \quad (43)$$

in which  $\mathcal{I}_h^q : Q_{n_q} \rightarrow Q_h$  is the interpolation operator for shear force resultant.  $\mathcal{W}_h : Q_h \rightarrow W_h$  is the orthogonal operator of  $\mathcal{W}$ , i.e.  $(\delta w_h, \mathcal{W}\mathbf{q}_h - \mathcal{W}_h\mathbf{q}_h) = 0$ .  $Q_{n_q} \subset Q$  is the polynomial space that has the identical dimensions with  $Q_h$ , i.e.  $\dim Q_{n_q} = \dim Q_h = 2n_q$ .

From Eq. (43), the LBB condition of Eq. (35) can be maintained while the first term on right-hand side of Eq. (43) is non-zero, which should be necessarily required that:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \quad (44)$$

where, obviously, leading the definitions of  $\mathcal{I}_h^q$  and  $Q_{n_q}$ , and the rank-nullity theorem, the left-hand side of Eq. (44) can be rewritten as:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}_h \mathcal{I}_h^q) &= \dim(Q_h \setminus \ker \mathcal{W}_h) \\ &= \dim Q_h - \dim(\ker \mathcal{W}_h) \\ &= \dim(\text{Im} \mathcal{W}_h) \\ &= \dim W_h = n_w \end{aligned} \quad (45)$$

In contrast, to evaluate the right-hand side of Eq. (44), we first introduce the definition of operator  $\mathcal{W}$  in (42),  $\ker \mathcal{W}$  can be represented as:

$$\ker \mathcal{W} = \ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q} \quad (46)$$

According to above equation, the space  $Q_{n_q} \setminus \ker \mathcal{W}$  can be expressed by two terms, one is regarding the kernel space of  $\mathcal{W}^\Omega$ , the other is regarding the kernel space of  $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$ , yields:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}) &= \dim(Q_{n_q} \setminus (\ker \mathcal{W}^\Omega \cap \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cup \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &= \dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) + \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) \\ &\quad - \dim((Q_{n_q} \setminus \ker \mathcal{W}^\Omega) \cap (Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q})) \end{aligned} \quad (47)$$

where the second and third equations of Eq. (47) is derived based on De Morgan's theorem[] and Grassmann's theorem[] respectively. For clarity, these two theorems are illustrated in Appendix B.

For the first term regarding the kernel space of  $\mathcal{W}^\Omega$ , in accordance with rank-nullity theorem[3], the dimension of this space is equal to the that of the image space of  $\mathcal{W}^\Omega$  as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^\Omega) = \dim Q_{n_q} - \dim(\ker \mathcal{W}^\Omega) = \dim(\text{Im} \mathcal{W}^\Omega) \quad (48)$$

Because of the  $Q_{n_q}$  is the second order polynomial tensor space with dimension of  $2n_q$  and the  $\mathcal{W}^\Omega$  is the divergence operator acting on  $Q_{n_q}$ , the polynomial in  $\text{Im} \mathcal{W}^\Omega$  is one order lower than polynomial order of  $Q_{n_q}$ . The following equation presents the detailed monomial basis of  $Q_{n_q}$ , and the kernel space of operator  $\mathcal{W}^\Omega$ :

$$Q_{n_q} = \text{span} \left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ x \end{pmatrix}, \begin{pmatrix} x \\ -y \end{pmatrix}, \dots, \begin{pmatrix} x \\ y \end{pmatrix}, \dots}_{\ker \mathcal{W}^\Omega} \right\} \quad (49)$$

To simplify the expression, we define two operators to implement the transformation between the order of a complete polynomial space, namely  $n_o$  and its monomial basis number, namely  $n_m$ :

$$n_o = [n_p] = \frac{\sqrt{1+8n_p}-3}{2}, \quad n_p = \langle n_o \rangle = \frac{(n_o+1)(n_o+2)}{2} \quad (50)$$

where  $\langle [n] \rangle = n$ . Therefore, the dimension of  $\text{Im} \mathcal{W}^\Omega$  can be estimated by:

$$\langle \lfloor [n_q] \rfloor - 1 \rangle \leq \dim(\text{Im} \mathcal{W}^\Omega) \leq \langle \lceil [n_q] \rceil - 1 \rangle \quad (51)$$

in which  $\lfloor \bullet \rfloor$  and  $\lceil \bullet \rceil$  are the floor and ceiling functions, respectively.

For the second term regarding the kernel space of  $\mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$ , leading the Lemma.1 listed in Appendix B, the dimension of  $Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}$  becomes:

$$\begin{aligned} \dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) &= \dim(Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}) \\ &= \dim Q_{n_q^\Gamma} - \dim(\ker \mathcal{W}^{\Gamma_q}) \\ &= \dim(\text{Im} \mathcal{W}^{\Gamma_q}) \end{aligned} \quad (52)$$

where  $Q_{n_q^\Gamma} \subset Q|_{\Gamma_q}$  is the polynomial space with the identical dimension of trace space  $\mathcal{T}^{\Gamma_q}(Q_h)$ , i.e.  $\dim Q_{n_q^\Gamma} = \dim \mathcal{T}^{\Gamma_q}(Q_h) = 2n_q^\Gamma$ . For instance, the detailed monomial basis of  $Q_{n_q^\Gamma}$  is listed as follows:

$$Q_{n_q^\Gamma} = \text{span} \left\{ \underbrace{\begin{pmatrix} n_2 \\ -n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ -n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ -n_1 s^2 \end{pmatrix}, \dots}_{\ker \mathcal{W}^{\Gamma_q}}, \underbrace{\begin{pmatrix} n_2 \\ n_1 \end{pmatrix}, \begin{pmatrix} n_2 s \\ n_1 s \end{pmatrix}, \begin{pmatrix} n_2 s^2 \\ n_1 s^2 \end{pmatrix}, \dots}_{Q_{n_q^\Gamma} \setminus \ker \mathcal{W}^{\Gamma_q}} \right\} \quad (53)$$



137 It is clear that, from Eq. (53), the dimension of second term is evaluated as:

$$\dim(Q_{n_q} \setminus \ker \mathcal{W}^{\Gamma_q} \mathcal{T}^{\Gamma_q}) = n_q^\Gamma \quad (54)$$

138 With the consideration of Eqs. (47), (51) and (54), the right-hand side of  
139 Eq. (44) becomes to:

$$\max(\langle \lfloor [n_q] \rfloor - 1 \rangle, n_q^\Gamma) \leq \dim(Q_{n_q} \setminus \ker \mathcal{W}) \leq \langle \lceil [n_q] \rceil - 1 \rangle + n_q^\Gamma \quad (55)$$

140 It is noted that the dimension of the trace space  $\mathcal{T}^{\Gamma_q}(Q_h)$  is generally iden-  
141 tical to the number of DOFs on  $\Gamma_q$ , i.g. for finite element approximation. So  
142 generally speaking,  $n_q^\Gamma \ll \langle \lfloor [n_q] \rfloor - 1 \rangle$ . To totally ensure the LBB condition of  
143 Eq. (35), considering Eqs. (44), (45) and (55), the  $n_w$  should be always less  
144 than the low bound of  $Q_{n_q} \setminus \ker \mathcal{W}$ 's dimension, i.e.:

$$n_w \leq \langle \lfloor [n_q] \rfloor - 1 \rangle \quad (56)$$

### 145 3.2. Estimator for $\beta_\gamma$

146 The behavior of shear strain locking in Eq. (36) is controlled by the coeffi-  
147 cient  $\beta_\gamma$ , where the shear strain locking contains two parts as follows:

$$\begin{aligned} & b(\delta \mathbf{q}_h, \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) = (\delta \mathbf{q}_h, \mathcal{Q} \mathbf{u}_h) - f_q(\delta \mathbf{q}_h) \\ & = \int_{\Omega} \delta \mathbf{q} \cdot (\nabla w - \boldsymbol{\varphi}) d\Omega - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} (w - \bar{w}) d\Gamma \\ & = 0 \end{aligned} \quad (57)$$

148 where the product operator herein stands for the tensors. And the operator  
149  $\mathcal{Q} : W \times \Phi \rightarrow Q$  is defined individually in the domain  $\Omega$  and on the boundary  
150  $\Gamma_w$  by:

$$\mathcal{Q} = \begin{cases} \mathcal{Q}^\Omega = \{\nabla, -\} & \text{in } \Omega \\ \mathcal{Q}^{\Gamma_w} \mathcal{T}^w, \mathcal{Q}^{\Gamma_w} = -\mathbf{n} & \text{on } \Gamma_w \end{cases} \quad (58)$$

151 in which, in accordance with trace theorem [],  $\mathcal{T}^w : W \times \Phi \rightarrow W|_{\Gamma_w}$  is the trace  
152 operator mapping displacement  $\mathbf{u}$  from domain  $\Omega$  to boundary  $\Gamma_w$ . Again,  
153 introducing the Theorem. 1, the  $\beta_\gamma$  can further be bounded by:

$$\beta_\gamma \leq \inf_{U' \subset W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u} \sup_{\mathbf{u}_h \in U'} \frac{\|\mathcal{Q} \mathbf{u}\|_Q}{\|\mathbf{u}\|_{W \times \Phi}} + O(h) \quad (59)$$

154 where  $\mathcal{I}_h^u : W_{n_w} \times \Phi_{n_\varphi} \rightarrow W_h \times \Phi_h$  is the interpolation operator for deflection  
155  $w$  and rotation  $\boldsymbol{\varphi}$ , and  $W_{n_w}$ ,  $\Phi_{n_\varphi}$  are the polynomial spaces that have the  
156 identical dimensions with  $W_h$ ,  $\Phi_h$  respectively, i.e.  $\dim W_{n_w} = \dim W_h = n_w$ ,  
157  $\dim \Phi_{n_\varphi} = \dim \Phi_h = 2n_\varphi$ .

158 To maintain the LBB condition of (36), i.e.  $\beta_\gamma$  is a coefficient independent  
159 with characterized length  $h$ , the dimensions of the relative spaces in Eq. (59)  
160 should satisfy the following relationship:

$$\dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) \leq \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \quad (60)$$

161 The left-hand side of Eq. (60) can be represented by the DOFs of approximated  
 162 shear force resultant:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}_h \mathcal{I}_h^u) &= \dim(W_h \times \Phi_h \setminus \ker \mathcal{Q}_h) \\
 &= \dim(W_h \times \Phi_h) - \dim(\ker \mathcal{Q}_h) \\
 &= \dim(\text{Im } \mathcal{Q}_h) \\
 &= \dim \mathcal{Q}_h = 2n_q
 \end{aligned} \tag{61}$$

163 On the other hand, with De Morgan's theorem and Grassmann's theorem [],  
 164 the right-hand side of Eq. (60) becomes:

$$\begin{aligned}
 &\dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus (\ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w})) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cup \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &= \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) + \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) \\
 &\quad - \dim((W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) \cap (W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}))
 \end{aligned} \tag{62}$$

165 where the relationship of  $\ker \mathcal{Q} = \ker \mathcal{Q}^\Omega \cap \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$  is used herein. Similarly,  
 166 this dimension of Eq. (62) is consisted of two terms, one relates to the kernel  
 167 space of  $\mathcal{Q}^\Omega$  in  $\Omega$ , the other relates to the kernel space of  $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$  on  $\Gamma_w$ .

168 The dimension of first term regarding the kernel space of  $\mathcal{Q}^\Omega$  can be revealed  
 169 by the detailed expression of monomial basis of  $W_{n_w} \times \Phi_{n_\varphi}$  as:

$$W_{n_w} \times \Phi_{n_\varphi} = \text{span} \left\{ \overbrace{\left( \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} x \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} y \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} x^2 \\ 2x \\ 0 \end{pmatrix}, \begin{pmatrix} xy \\ y \\ x \end{pmatrix}, \begin{pmatrix} y^2 \\ 0 \\ 2y \end{pmatrix}, \dots \right)}^{\ker \mathcal{Q}^\Omega}, \underbrace{\left( \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ x \end{pmatrix}, \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ y \end{pmatrix}, \dots \right)}_{W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega} \right\} \tag{63}$$

170 Obviously, the monomial basis of deflection  $w$  is one order higher than that of  
 171 rotation in  $\ker \mathcal{Q}^\Omega$ . So the dimension of  $\ker \mathcal{Q}^\Omega$  can be evaluated by:

$$\dim \ker \mathcal{Q}^\Omega = \min(n_w, n_s) \tag{64}$$

172 with

$$\langle \lfloor n_\varphi \rfloor + 1 \rangle \leq n_s \leq \langle \lceil n_\varphi \rceil + 1 \rangle \tag{65}$$

173 where  $n_w$  presents the number of monomial basis for  $w$ .  $n_s$  stands for the  
 174 monomial basis number with one order higher than  $\Phi_{n_\varphi}$ 's monomial basis. Then,  
 175 the dimension of  $W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega$  can be evaluated by:

$$\begin{aligned}
 \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^\Omega) &= \dim(W_{n_w} \times \Phi_{n_\varphi}) - \dim(\ker \mathcal{Q}^\Omega) \\
 &= n_w + 2n_\varphi - \min(n_w, n_s)
 \end{aligned} \tag{66}$$

176 To further calculate the dimension of the second term regarding the kernel  
 177 space of  $\mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}$ , we first suppose the trace space  $\mathcal{T}^{\Gamma_w}(W_h \times \Phi_h)$  and its  
 178 corresponding polynomial space  $W_{n_w}^{\Gamma}$ ,  $\dim W_{n_w}^{\Gamma} = \dim \mathcal{T}^{\Gamma_w}(W_h \times \Phi_h) = n_w^{\Gamma}$ .  
 179 Using the Lemma.1 in Appendix B, we have:

$$\begin{aligned} \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}^{\Gamma_w} \mathcal{T}^{\Gamma_w}) &= \dim(W_{n_w}^{\Gamma} \setminus \ker \mathcal{Q}^{\Gamma_w}) \\ &= \dim W_{n_w}^{\Gamma} - \underbrace{\dim(\ker \mathcal{Q}^{\Gamma_w})}_0 \\ &= \dim W_{n_w}^{\Gamma} = n_w^{\Gamma} \end{aligned} \quad (67)$$

180 where, it can reveal from the expression of  $\mathcal{Q}^{\Gamma_w}$  in Eq.(58) that, the space  
 181  $\ker \mathcal{Q}^{\Gamma_w}$  is empty, i.e  $\dim(\ker \mathcal{Q}^{\Gamma_w}) = 0$ .

182 Consequently, substituting Eqs.(66) and (67) into Eq.(62), the right-hand  
 183 side of Eq.(60) can be rewritten as:

$$\begin{aligned} \max(n_w + 2n_\varphi - \min(n_w, n_s), n_w^{\Gamma}) &\leq \dim(W_{n_w} \times \Phi_{n_\varphi} \setminus \ker \mathcal{Q}) \\ &\leq n_w + 2n_\varphi - \min(n_w, n_s) + n_w^{\Gamma} \end{aligned} \quad (68)$$

184 It is clear that  $n_w^{\Gamma} \ll n_w + 2n_\varphi - \min(n_w, n_s)$  FIXME

$$n_q \leq n_\varphi \quad (69)$$

$$n_w^{\text{opt}} = \langle \lfloor n_q \rfloor - 1 \rangle, \quad n_q^{\text{opt}} = n_\varphi \quad (70)$$

186 **Remark 3.** It should be noted that, the proposed estimator is considered for  
 187 the entire domain and all the nodal discretization should be so well-posed that  
 188 the monomial basis number of equivalent polynomial space is equal to the degree  
 189 of freedoms for approximation space, i.g  $\dim W_{n_w} = \dim W_h = n_w$ . While  
 190 a local domain contains a too dense nodal distribution than others, the nodal  
 191 discretization satisfied the estimator will still meet the locking issue in local.

192 **Remark 4.** The shear force locking will influence the shear strain locking.

## 193 4. Shear-locking-free FE-Meshfree formulation

### 194 4.1. Reproducing kernel meshfree approximation

$$\Psi_I(\mathbf{x}) = \mathbf{c}^T(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (71)$$

$$\phi(\mathbf{x}_I - \mathbf{x}) = \phi(r_{1I}) \phi(r_{2I}), \quad r_{iI} = \frac{|\mathbf{x}_{iI} - \mathbf{x}_i|}{s_i} \quad (72)$$

$$\phi(r) = \frac{1}{3!} \begin{cases} (2-2r)^3 - 4(1-2r)^3 & r \leq \frac{1}{2} \\ (2-2r)^3 & \frac{1}{2} < r \leq 1 \\ 0 & r > 1 \end{cases} \quad (73)$$

197

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I) = \mathbf{p}(\mathbf{x}) \quad (74)$$

198 or a shifted form

$$\sum_{I=1}^n \Psi_I(\mathbf{x}) \mathbf{p}(\mathbf{x}_I - \mathbf{x}) = \mathbf{p}(\mathbf{0}) \quad (75)$$

199

$$\mathbf{c}(\mathbf{x}_I - \mathbf{x}) = \mathbf{A}^{-1}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}(\mathbf{0}) \quad (76)$$

200 with

$$\mathbf{A}(\mathbf{x}_I - \mathbf{x}) = \sum_{I=1}^n \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \mathbf{p}^T(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (77)$$

201

$$\Psi_I(\mathbf{x}) = \mathbf{p}^T(\mathbf{0}) \mathbf{A}^{-1} \mathbf{p}(\mathbf{x}_I - \mathbf{x}) \phi(\mathbf{x}_I - \mathbf{x}) \quad (78)$$

202 *4.2. Eigenvalue test for checking LBB coefficients(should be fixed)*

203 To verify the optimal constraint ratios proposed in Eq. (70), the eigenvalue  
 204 test of LBB coefficients [4] in a square domain, shown in Figure.??, is intro-  
 205 duced herein, where all the boundaries of this domain are clamped. In order  
 206 to adjust the constraint ratio more flexibly, the reproducing kernel meshfree  
 207 approximation are used to discretize the deflection  $w$  and rotation  $\varphi$ , yields:

$$N_I^w := \Psi_I, N_K^\varphi := \Psi_K^\varphi, N_R^q := N_R \quad (79)$$

208 while, as shown in Figure.??, the meshfree nodes for  $w$  and  $\varphi$  are different.

209 To meet the variational consistency of the weak formulation with meshfree  
 210 approximation, the reproducing kernel gradient smoothing integration frame-  
 211 work (RKGSi) [5] is used herein. Accordingly, the Mindlin plate version of  
 212 RKGSi framework is presented in Appendix C.

213 *4.3. Shear-locking-free formulations with optimal constraint ratio*

$$N_I^w := \Psi_I, N_K^\varphi := N_K, N_R^q := N_R \quad (80)$$

214 FIXME

215 **Remark 5.** *The discretization of Eq.*216 **5. Numerical examples**217 *5.1. Square plate subjected non-uniform transverse load*

218 Consider a square plate with the clamped boundary, as showing in Figure.  
 219 , subjecting a non-uniform transverse load  $\bar{p}$ :

$$\bar{p} = \frac{Eh^3}{12(1-\nu^2)} \left( \begin{array}{l} 12y(y-1)(5x^2-5x+1)(2y^2(y-1)^2+x(x-1)(5y^2-5y+1)) \\ +12x(x-1)(5y^2-5y+1)(2x^2(x-1)^2+y(y-1)(5x^2-5x+1)) \end{array} \right) \quad (81)$$

220 In this circumstance, the exact solution of the deflection and rotation are listed  
 221 as follows:

$$w = \frac{1}{3}x^3(x-1)^3y^3(y-1)^3 - \frac{2h^2}{5(1-\nu)} \left( \frac{y^3(y-1)^3x(x-1)(5x^2-5x+1)+}{x^3(x-1)^3y(y-1)(5y^2-5y+1)} \right) \quad (82)$$

$$\begin{cases} \varphi_1 = y^3(y-1)^3x^2(x-1)^2(2x-1) \\ \varphi_2 = x^3(x-1)^3y^2(y-1)^2(2y-1) \end{cases} \quad (83)$$

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## 227 Appendix A. Complete-polynomial-based estimator for LBB coefficient 228

229 This appendix presents the detailed derivation of the LBB coefficient esti-  
230 mator used in this work. This estimator provides a quantitative measure of  
231 the stability of mixed finite element formulations just by counting the degree  
232 of freedoms (DOFs) of the approximation spaces. The main route of derivation  
233 follows the path proposed in Ref. [2]. Unlike the consideration of volumetric  
234 locking in [2], here we focus on the general LBB condition for any locking is-  
235 sue. Consider the following constraint servicing to a saddle-point problem: find  
236  $u \in U$  such that

$$b(p, u) = g(p), \quad \forall p \in P \quad (\text{A.1})$$

237 where  $b : U \times P \rightarrow \mathbb{R}$ ,  $g : P \rightarrow \mathbb{R}$  are bilinear and linear forms, respectively. For  
238 an any fixed  $p \in P$ , there exists a linear operator  $\mathcal{P} : U \rightarrow P$  satisfying:

$$b(p, u) = (p, \mathcal{P}u) \quad (\text{A.2})$$

239 where  $(\bullet, \bullet) : P \times P \rightarrow \mathbb{R}$  is the inner product in  $P$ .

240 By introducing the discrete spaces  $U_h \subset U$  and  $P_h \subset P$ , the constraint of  
241 Eq. (A.1) is approximated as: find  $u_h \in U_h$  such that

$$b(p_h, u_h) = g(p_h), \quad \forall p_h \in P_h \quad (\text{A.3})$$

242 and  $\mathcal{P}$ 's orthogonal projection operator regarding to  $P_h$ , namely  $\mathcal{P}_h : U_h \rightarrow P_h$ ,  
243 is defined by

$$(p_h, \mathcal{P}u_h - \mathcal{P}_h u_h) = 0 \Rightarrow (p_h, \mathcal{P}u_h) = (p_h, \mathcal{P}_h u_h), \quad \forall p_h \in P_h \quad (\text{A.4})$$

244 To obtain the satisfaction solution of saddle-point problems with constraint  
245 of Eq. (A.3), the well-known LBB condition is required, i.e., there exists a  
246 positive constant  $\beta > 0$ , namely LBB coefficient, such that

$$\inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|b(p_h, u_h)|}{\|p\|_P \|u\|_U} \geq \beta \quad (\text{A.5})$$

247 where  $\|\bullet\|_U$  and  $\|\bullet\|_P$  are the norms in spaces  $U$  and  $P$ , respectively. To  
248 further estimate the LBB coefficient  $\beta$  by the degree of freedom of  $U_h$  and  $P_h$ ,  
249 i.e.  $n_u = \dim U_h$  and  $n_p = \dim P_h$ , we introduce the following theorem.

250 **Theorem 1.** (Completed-polynomial-based estimator for LBB coefficient) Sup-  
251 pose that  $\mathcal{P} : U \rightarrow P$  is the linear operator representing the bilinear form  $b$   
252 in LBB condition and  $\mathcal{P}_h : U_h \rightarrow P_h$  is the corresponding orthogonal projec-  
253 tion operator regarding to  $P_h$ ,  $U_{n_u} \subset U$  is a complete polynomial space with  
254  $\dim U_{n_u} = n_u$ . The LBB coefficient  $\beta$  can be estimated by:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}u\|_P}{\|u\|_U} + O(h) \quad (\text{A.6})$$

255 where  $\mathcal{I}_h : U_{n_u} \rightarrow U_h$  is the interpolation operator with the characteristic size  
256 of the discretization  $h$ .

257 PROOF. Firstly, with the definition of  $\mathcal{P}$  and  $\mathcal{P}_h$  in Eqs. (A.2) and (A.4) re-  
 258 spectively, the LBB condition can be rewritten as:

$$\begin{aligned} \beta &\leq \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}u_h)|}{\|p_h\|_P \|u_h\|_U} = \inf_{p_h \in P_h} \sup_{u_h \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \\ &\leq \inf_{p_h \in \text{Im} \mathcal{P}_h} \sup_{u \in U_h} \frac{|(p_h, \mathcal{P}_h u_h)|}{\|p_h\|_P \|u_h\|_U} \end{aligned} \quad (\text{A.7})$$

259 FIXME: the definition of image space can be canceled. where  $\text{Im} \mathcal{P}_h$  is the  
 260 image space of operator  $\mathcal{P}_h$ , i.e.  $\text{Im} \mathcal{P}_h := \{p_h \in P_h \mid \exists u_h \in U_h, \mathcal{P}_h u_h = p_h\}$ ,  
 261  $\text{Im} \mathcal{P}_h \subset P_h$ . Since both  $p_h$  and  $\mathcal{P}_h u_h$  belong to  $\text{Im} \mathcal{P}_h$ , using Cauchy-Schwarz  
 262 inequality, we can further obtain that:

$$|(\mathcal{P}_h u_h, p_h)| \leq \|\mathcal{P}_h u_h\|_P \|p_h\|_P \quad (\text{A.8})$$

263 For a given  $p_h$ , with introduction of a space  $U' := \{u_h \in U_h \mid \mathcal{P}_h u_h = p_h\}$   
 264 regarding to  $p_h$ , the equality in Eq. (A.7) holds when  $p_h = \mathcal{P}_h u_h$ , yields:

$$|(\mathcal{P}_h u_h, p_h)| = \|\mathcal{P}_h u_h\|_P \|p_h\|_P, \quad \forall u_h \in U' \quad (\text{A.9})$$

265 and, obviously,  $U' \subseteq U_h \setminus \ker \mathcal{P}_h$ ,  $\ker \mathcal{P}_h$  is the kernel space of operator  $\mathcal{P}_h$ , i.e.  
 266  $\ker \mathcal{P}_h := \{u_h \in U_h \mid \mathcal{P}_h u_h = 0\}$ . Substituting Eq. (A.9) into Eq. (A.7) leads  
 267 to:

$$\beta \leq \inf_{U' \subset U_h \setminus \ker \mathcal{P}_h} \sup_{u_h \in U'} \frac{\|\mathcal{P}_h u_h\|_P}{\|u_h\|_U} \quad (\text{A.10})$$

268 Secondly, the complete polynomial space  $V_{n_u} \subset U$  with identical dimension  
 269 to  $U_h$  is introduced herein to further estimate the  $\beta$ , where  $n_u$  is the dimension  
 270 of  $U_h$ , i.e.  $n_u = \dim U_h = \dim V_{n_u}$ . At this circumstance, for any  $u \in U_{n_u}$ ,  
 271 there exists an interpolation operator  $\mathcal{I}_h : U \rightarrow U_h$  such that  $u_h = \mathcal{I}_h u$ . Thus,  
 272 Eq. (A.10) can be rewritten as:

$$\beta \leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \quad (\text{A.11})$$

273 In accordance with the triangular inequality [6] and the relationship of Eq.  
 274 (A.4), we have:

$$\begin{aligned} \|\mathcal{P}_h \mathcal{I}_h u\|_P &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P}_h \mathcal{I}_h u, p_h)|}{\|p_h\|_P} = \sup_{p_h \in P_h} \frac{|(\mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &\leq \sup_{p_h \in P_h} \frac{|(\mathcal{P} u, p_h)| + |(\mathcal{P} u - \mathcal{P} \mathcal{I}_h u, p_h)|}{\|p_h\|_P} \\ &= \|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P \end{aligned} \quad (\text{A.12})$$

275 where the last term is the interpolation error of space  $U_h$  with respect to operator  
 276  $\mathcal{P}$ . Generally, particularly for finite element approximations, the interpolation  
 277 error is proportional to the characteristic size of the discretization  $h$  [6]:

$$\|\mathcal{P}(u - \mathcal{I}_h u)\|_P \leq Ch^k |u|_{H_k} \quad (\text{A.13})$$

278 in which  $C$  is a positive constant independent of  $h$ ,  $k$  is the convergence order  
 279 related to the operators  $\mathcal{I}_h$  and  $\mathcal{P}$ . With the combination of Eqs. (A.11), (A.12)  
 280 and (A.13), the final estimator of LBB coefficient  $\beta$  in Eq. (1) can be proved:

$$\begin{aligned}
 \beta &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P}_h \mathcal{I}_h u\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P + \|\mathcal{P}(u - \mathcal{I}_h u)\|_P}{\|\mathcal{I}_h u\|_U} \\
 &\leq \inf_{U' \subset U_{n_u} \setminus \ker \mathcal{P}_h \mathcal{I}_h} \sup_{u \in U'} \frac{\|\mathcal{P} u\|_P}{\|u\|_U} + O(h)
 \end{aligned} \tag{A.14}$$

281        FIXME:use the remark env It should be noted that Theorem 1 inherently  
 282 embeds the numbers of DOFs of approximation spaces  $U_h$  and  $P_h$ , while  $n_u =$   
 283  $\dim U_h$  and  $n_p = \dim \text{Im} \mathcal{P}_h = \dim(U_h \setminus \ker \mathcal{P}_h \mathcal{I}_h)$ , thus it is easy to justify the  
 284 LBB condition by just counting the DOFs of approximation spaces.



## 285 Appendix B. Preliminaries for functional analysis

286 In the interest of conciseness, kernel space

$$\ker \mathcal{P} := \{u \in U \mid \mathcal{P}u = 0\} \quad (\text{B.1})$$

287 image space

$$\text{Im} \mathcal{P} := \{p \in P \mid \exists u \in U, p = \mathcal{P}u\} \quad (\text{B.2})$$

288 It is clear that  $\ker \mathcal{P} \subset U$  and  $\text{Im} \mathcal{P} \in P$ .

289 **FIXME:** Trace's theorem De Morgan's theorem Grassmann's theorem Rank-  
290 nullity theorem

291 **Lemma 1.** *Suppose that  $\mathcal{T} : U \rightarrow U|_{\Gamma}$  is the trace operator of  $U$ ,  $U|_{\Gamma}$  is  $U$ 's  
292 trace space on the boundary  $\Gamma$ . Let  $\mathcal{L}$  is an operator acting on  $U|_{\Gamma}$ , then we  
293 have:*

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.3})$$

294 **PROOF.** In accordance with rank-nullity theorem, we have the following rela-  
295 tionship:

$$\dim U = \dim(\text{Im} \mathcal{L}\mathcal{T}) + \dim(\ker \mathcal{L}\mathcal{T}) \quad (\text{B.4})$$

296 or

$$\dim(U \setminus \ker \mathcal{L}\mathcal{T}) = \dim U - \dim(\ker \mathcal{L}\mathcal{T}) = \dim(\text{Im} \mathcal{L}\mathcal{T}) \quad (\text{B.5})$$

297 in which, according to the definition of the image space  $\text{Im} \mathcal{L}\mathcal{T}$ , it is clear that:

$$\text{Im} \mathcal{L}\mathcal{T} = \{\mathcal{L}\mathcal{T}u \mid u \in U\} = \{\mathcal{L}(\mathcal{T}u) \mid \mathcal{T}u \in U|_{\Gamma}\} = \text{Im} \mathcal{L} \quad (\text{B.6})$$

298 Subsequently, the dimension of image space for operator  $\mathcal{L}$ , i.e.  $\dim(\text{Im} \mathcal{L})$ ,  
299 can be evaluated by introducing rank-nullity theorem again as follows:

$$\dim U|_{\Gamma} = \dim(\text{Im} \mathcal{L}) + \dim(\ker \mathcal{L}) \quad (\text{B.7})$$

300 or

$$\dim(\text{Im} \mathcal{L}) = \dim U|_{\Gamma} - \dim(\ker \mathcal{L}) = \dim(U|_{\Gamma} \setminus \ker \mathcal{L}) \quad (\text{B.8})$$

301 With the combination of Eqs. (B.5), (B.6) and (B.8), the relationship of Eq.  
302 (B.3) can be proofed.

303 **Appendix C. Reproducing kernel smoothed gradient method for Mindlin**  
 304 **plate formulation**

$$\begin{aligned}
 \Pi(w, \boldsymbol{\varphi}, \mathbf{q}, \mathbf{m}) = & - \int_{\Omega} \frac{1}{2} \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Omega} \frac{1}{2} \frac{\mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega \\
 & - \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma - \int_{\Gamma_w} \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
 & - \int_{\Gamma_m} \boldsymbol{\varphi} \cdot (\mathbf{m} \cdot \mathbf{n} - \bar{\mathbf{m}}_n) d\Gamma - \int_{\Omega} \boldsymbol{\varphi} \cdot (\nabla \cdot \mathbf{m} - \mathbf{q}) d\Omega \\
 & + \int_{\Gamma_q} w(\mathbf{q} \cdot \mathbf{n} - \bar{\mathbf{q}}_n) d\Gamma - \int_{\Omega} w(\nabla \cdot \mathbf{q} + \bar{p}) d\Omega
 \end{aligned} \tag{C.1}$$

$$\begin{aligned}
 a_m(\delta \mathbf{m}, \mathbf{m}) + b_m(\delta \mathbf{m}, \boldsymbol{\varphi}) &= f_m(\delta \mathbf{m}), \quad \forall \delta \mathbf{m} \in M \\
 a_q(\delta \mathbf{q}, \mathbf{q}) + b_q(\delta \mathbf{q}, \mathbf{u}) &= f_q(\delta \mathbf{q}), \quad \forall \delta \mathbf{q} \in Q \\
 b_w(\delta w, \mathbf{q}) &= f_w(\delta w), \quad \forall \delta w \in W \\
 b_m(\delta \boldsymbol{\varphi}, \mathbf{m}) + b_{\varphi}(\delta \boldsymbol{\varphi}, \mathbf{q}) &= f_{\varphi}(\delta \boldsymbol{\varphi}), \quad \forall \delta \boldsymbol{\varphi} \in \Phi
 \end{aligned} \tag{C.2}$$

$$a_m(\delta \mathbf{m}, \mathbf{m}) = - \int_{\Omega} \delta \mathbf{m} : \frac{12}{t^3} \mathbf{D}^{-1} : \mathbf{m} d\Omega \tag{C.3}$$

$$b_m(\mathbf{m}, \boldsymbol{\varphi}) = - \int_{\Gamma_m} \mathbf{n} \cdot \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega \tag{C.4}$$

$$f_m(\boldsymbol{\varphi}, \mathbf{q}) = \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \tag{C.5}$$

$$\tag{C.6}$$

$$\begin{aligned}
 - \int_{\Omega} \delta \mathbf{m} : \mathbf{D}^{-1} : \mathbf{m} d\Omega - \int_{\Gamma_m} \mathbf{n} \cdot \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Gamma + \int_{\Omega} \nabla \delta \mathbf{m} \cdot \boldsymbol{\varphi} d\Omega &= \int_{\Gamma_{\varphi}} \mathbf{n} \cdot \delta \mathbf{m} \cdot \bar{\boldsymbol{\varphi}} d\Gamma \\
 - \int_{\Omega} \frac{\delta \mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega + \int_{\Gamma_q} \delta \mathbf{q} \cdot \mathbf{n} w d\Gamma - \int_{\Omega} \nabla \cdot \delta \mathbf{q} w d\Omega + \int_{\Omega} \delta \mathbf{q} \cdot \boldsymbol{\varphi} d\Omega &= - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma
 \end{aligned} \tag{C.7}$$

$$\begin{aligned}
 - \int_{\Omega} \frac{\delta \mathbf{q} \cdot \mathbf{q}}{kGh} d\Omega + \int_{\Gamma_q} \delta \mathbf{q} \cdot \mathbf{n} w d\Gamma - \int_{\Omega} \nabla \cdot \delta \mathbf{q} w d\Omega + \int_{\Omega} \delta \mathbf{q} \cdot \boldsymbol{\varphi} d\Omega &= - \int_{\Gamma_w} \delta \mathbf{q} \cdot \mathbf{n} \bar{w} d\Gamma \\
 \int_{\Gamma_q} \delta w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta w \nabla \cdot \mathbf{q} d\Omega &= \int_{\Gamma_q} \delta w \bar{\mathbf{q}}_n d\Gamma - \int_{\Omega} \delta w \bar{p} d\Omega
 \end{aligned} \tag{C.8}$$

$$\int_{\Gamma_q} \delta w \mathbf{q} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta w \nabla \cdot \mathbf{q} d\Omega = \int_{\Gamma_q} \delta w \bar{\mathbf{q}}_n d\Gamma - \int_{\Omega} \delta w \bar{p} d\Omega \tag{C.9}$$

$$\begin{aligned}
 - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \mathbf{n} d\Gamma - \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{m} \cdot \nabla d\Omega + \int_{\Omega} \delta \boldsymbol{\varphi} \cdot \mathbf{q} d\Omega &= - \int_{\Gamma_m} \delta \boldsymbol{\varphi} \cdot \bar{\mathbf{m}}_n d\Gamma \\
 - \int_{\Omega_C} \delta m_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} m_{\gamma\eta} d\Omega - \int_{\Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma + \int_{\Omega_C} \delta M_{\alpha\beta, \beta} \varphi_{\alpha} d\Omega + \int_{\Gamma_{\varphi} \cap \Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma &= \int_{\Gamma_{\varphi} \cap \Gamma_C} M_{\alpha\beta} n_{\beta} \bar{\varphi}_{\alpha} d\Gamma
 \end{aligned} \tag{C.10}$$

$$\begin{aligned}
 - \int_{\Omega_C} \delta m_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} m_{\gamma\eta} d\Omega - \int_{\Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma + \int_{\Omega_C} \delta M_{\alpha\beta, \beta} \varphi_{\alpha} d\Omega + \int_{\Gamma_{\varphi} \cap \Gamma_C} \delta M_{\alpha\beta} n_{\beta} \varphi_{\alpha} d\Gamma &= \int_{\Gamma_{\varphi} \cap \Gamma_C} M_{\alpha\beta} n_{\beta} \bar{\varphi}_{\alpha} d\Gamma \\
 &\tag{C.11}
 \end{aligned}$$

$$m_{\alpha\beta h}(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{a}_{\alpha\beta C} \quad \text{in } \Omega_C \quad (\text{C.12})$$

$$\mathbf{a}_{\alpha\beta C} = D_{\alpha\beta\eta\gamma} \mathbf{G}_C^{-1} \left( \sum_{K=1}^{n_\varphi} (\tilde{\mathbf{g}}_{\eta K} + \hat{\mathbf{g}}_{\eta K}) \varphi_{\gamma K} + \bar{\mathbf{g}}_{\gamma\eta} \right) \quad (\text{C.13})$$

$$\tilde{\mathbf{g}}_{\alpha K} = - \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\alpha d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\alpha} N_K^\varphi d\Omega \quad (\text{C.14})$$

$$\bar{\mathbf{g}}_{\alpha K} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\alpha d\Gamma \quad (\text{C.15})$$

$$\hat{\mathbf{g}}_{\alpha\beta} = \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} n_\beta \bar{\varphi}_\alpha d\Gamma \quad (\text{C.16})$$

$$(\text{C.17})$$

$$\mathbf{m}_h = \tilde{\mathbf{m}}_h + \hat{\mathbf{m}}_h + \bar{\mathbf{m}}_h \quad (\text{C.18})$$

$$\mathbf{m}(\varphi_h) = \mathbf{D} : (\tilde{\boldsymbol{\kappa}}(\varphi_h) + \hat{\boldsymbol{\kappa}}_h + \bar{\boldsymbol{\kappa}}_{\gamma\eta}) \quad (\text{C.19})$$

$$\tilde{m}_{\alpha\beta h} = -D_{\alpha\beta\eta\gamma} (\tilde{\kappa}_{\gamma\eta h} + \bar{\kappa}_{\gamma\eta} + \hat{\kappa}_{\gamma\eta}) \quad (\text{C.20})$$

with

$$\tilde{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left( \tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \tilde{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.21})$$

$$\bar{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\frac{1}{2} \sum_{K=1}^{n_\varphi} \left( \bar{N}_{K,\alpha}^\varphi(\mathbf{x}) \varphi_{\beta K} + \bar{N}_{K,\beta}^\varphi(\mathbf{x}) \varphi_{\alpha K} \right) \quad (\text{C.22})$$

$$\hat{\kappa}_{\alpha\beta h}(\mathbf{x}) = -\mathbf{G}_C^{-1} \hat{\mathbf{g}}_{\alpha\beta} \quad (\text{C.23})$$

$$\tilde{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\alpha K} \quad (\text{C.24})$$

$$\bar{N}_{K,\alpha}^\varphi(\mathbf{x}) = \mathbf{p}^T(\mathbf{x}) \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\alpha K} \quad (\text{C.25})$$

$$\begin{aligned} & b_m(\delta\varphi_h, \mathbf{m}_h) \\ &= - \int_{\Gamma_m} \delta\varphi_h \cdot \mathbf{m}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \mathbf{m}_h \cdot \nabla d\Omega \\ &= - \int_{\Gamma} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma + \int_{\Omega} \delta m_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_\varphi} \delta\varphi_{\alpha h} m_{\alpha\beta h} n_\beta d\Gamma \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left( - \int_{\Gamma_C} \mathbf{p} N_K^\varphi n_\beta d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^\varphi d\Omega + \int_{\Gamma_C \cap \Gamma_\varphi} \mathbf{p} N_K^\varphi n_\beta d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} ((\tilde{\mathbf{g}}_{\eta L} + \bar{\mathbf{g}}_{\eta L}) \varphi_{\gamma L} + \hat{\mathbf{g}}_{\gamma\eta}) \\ &= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_\varphi} \delta\varphi_{\alpha K} \left( \int_{\Omega_C} \tilde{\mathbf{g}}_{\beta K}^T \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \right) \varphi_{\gamma L} \end{aligned} \quad (\text{C.26})$$

318

$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \underbrace{\int_{\Gamma} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega}_{\int_{\Omega} \delta\tilde{\kappa} : \frac{t^3}{12} \mathbf{D} : \tilde{\kappa} d\Omega} + \int_{\Gamma_{\varphi}} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma \quad (\text{C.27})
\end{aligned}$$

319

$$b_m(\delta\varphi_h, \hat{\mathbf{m}}_h) = \tilde{b}_{\varphi}(\varphi_h, \tilde{\mathbf{m}}(\delta\varphi_h)) + \tilde{b}_{\varphi}(\delta\varphi_h, \hat{\mathbf{m}}(\varphi_h)) \quad (\text{C.28})$$

320

$$b_m(\delta\varphi_h, \bar{\mathbf{m}}_h) = \hat{f} + \bar{f} \quad (\text{C.29})$$

321

$$\begin{aligned}
& b_m(\delta\varphi_h, \tilde{\mathbf{m}}_h) \\
&= - \int_{\Gamma_m} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \mathbf{n} d\Gamma + \int_{\Omega} \delta\varphi_h \cdot \tilde{\mathbf{m}}_h \cdot \nabla d\Omega \\
&= - \int_{\Gamma} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma + \int_{\Omega} \delta\tilde{m}_{\alpha\beta h, \beta} \varphi_{\alpha h} d\Omega + \int_{\Gamma_{\varphi}} \delta\varphi_{\alpha h} \tilde{m}_{\alpha\beta h} n_{\beta} d\Gamma \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} \left( - \int_{\Gamma_C} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma + \int_{\Omega_C} \mathbf{p}_{,\beta} N_K^{\varphi} d\Omega + \int_{\Gamma_C \cap \Gamma_{\varphi}} \mathbf{p} N_K^{\varphi} n_{\beta} d\Gamma \right) D_{\alpha\beta\gamma\eta} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \sum_{K,L=1}^{n_{\varphi}} \delta\varphi_{\alpha K} (\tilde{\mathbf{g}}_{\beta K} + \hat{\mathbf{g}}_{\beta K}) \mathbf{G}_C^{-1} \int_{\Omega_C} \mathbf{p} D_{\alpha\beta\gamma\eta} \mathbf{p}^T d\Omega \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \\
&= \sum_{C=1}^{n_e} \left( \underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\tilde{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&+ \sum_{C=1}^{n_e} \left( \underbrace{\int_{\Omega_C} \sum_{K=1}^{n_{\varphi}} \mathbf{p}^T \mathbf{G}_C^{-1} \bar{\mathbf{g}}_{\beta K} \delta\varphi_{\alpha K} D_{\alpha\beta\gamma\eta}}_{\delta\bar{\kappa}_{\alpha\beta}} \underbrace{\sum_{L=1}^{n_{\varphi}} \mathbf{G}_C^{-1} \tilde{\mathbf{g}}_{\eta L} \varphi_{\gamma L}}_{\tilde{\kappa}_{\gamma\eta}} \right) \\
&= a_{\varphi}(\tilde{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) + a_{\varphi}(\hat{\kappa}(\delta\varphi), \tilde{\kappa}(\varphi)) \quad (\text{C.30})
\end{aligned}$$

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